LeafEngines™

Data Integrity at the Edge

Technical White Paper

20 Environmental Intelligence Endpoints | Offline-First Architecture | MCP Native

April 2026 | Version 2.0

Now Available On:

Node-RED Flows Library

npm Registry

GitHub — Source & Documentation

soilsidekickpro.com/podcast

1. Executive Summary

LeafEngines is a 20-endpoint environmental intelligence API designed for data integrity at the edge. Built on an offline-first architecture with hierarchical caching, the platform ensures agricultural and environmental data remains accurate, traceable, and available regardless of connectivity.

This white paper details the technical architecture, edge inference pipeline, Model Context Protocol (MCP) integration, and the API surface that positions LeafEngines as the industry standard for environmental data integrity.

2. Architecture: Offline-First Data Integrity

2.1 Hierarchical Cache System

LeafEngines implements a 4-level hierarchical cache to maintain data freshness while operating in connectivity-constrained environments:

- L1 — In-Memory (hot path): Sub-millisecond access for active queries
- L2 — IndexedDB (device-local): Persistent storage surviving app restarts
- L3 — Edge Function Cache: CDN-level caching at Supabase edge nodes
- L4 — Origin (PostgreSQL + PostGIS): Authoritative source with FIPS-indexed geospatial queries

2.2 Sync Queue with Exactly-Once Delivery

Offline mutations are queued locally and replayed upon reconnection using idempotency keys. The sync engine guarantees exactly-once delivery semantics, preventing duplicate soil analyses, duplicate carbon credit calculations, or duplicate field boundary updates. Conflict resolution follows a last-write-wins strategy with server-side timestamp validation.

2.3 Data Provenance & Uncertainty Quantification

Every API response includes provenance metadata: data source, retrieval timestamp, cache level served from, and staleness indicator. Soil and environmental data carry uncertainty bounds derived from USDA SSURGO sampling density and interpolation methods.

3. Edge Inference Pipeline

3.1 Model Architecture

LeafEngines deploys Gemma 4 26B MoE (Mixture of Experts) for on-device inference. With 26 billion total parameters but only 3.8 billion active per forward pass, the model delivers near-cloud accuracy at edge-viable compute budgets.

Runtime: ONNX Runtime with WebGPU/WebNN acceleration

Quantization: INT4 via TurboQuant, reducing model size by ~75% with <2% accuracy degradation

Fallback: WASM SIMD for devices without GPU support

3.2 Smart LLM Selection

The inference pipeline automatically selects the optimal execution path based on device capabilities, network status, and query complexity. GPU-capable devices run local inference; constrained devices route to edge functions; complex multi-modal queries escalate to cloud endpoints.

4. Model Context Protocol (MCP) Integration

LeafEngines implements the Model Context Protocol over Streamable HTTP, enabling autonomous AI agent discovery and invocation of all 20 endpoints without manual integration.

4.1 Agent Discovery

MCP-compatible agents (Claude, custom agent frameworks) can discover LeafEngines tools via the standard /.well-known/ai-plugin.json manifest. Each tool exposes typed input/output schemas, enabling agents to compose multi-step environmental intelligence workflows autonomously.

4.2 Context Modes

- Stateless: Single request/response (default)
- Session: Multi-turn with KV-cache hints for conversational workflows
- Batch: Parallel tool invocation for bulk processing

5. API Surface: 20 Endpoints Across 4 Tiers

Tier	Price	Rate Limit	Endpoints
Free (Sandbox)	$0	100 req/hr	get-soil-data, county-lookup
Starter ($149/mo)	$149/mo	10K req/mo	water-quality, dynamic-care, safe-identification, seasonal-planning, beginner-guidance, sensor-data-quality, smart-report-summary, multi-parameter-planting-calendar
Pro ($499/mo)	$499/mo	50K req/mo	agricultural-intelligence, carbon-credit-calculator, generate-vrt-prescription, environmental-impact-engine, live-agricultural-data, plant-id-comparison, isobus-task
Enterprise ($1,500/mo)	$1,500/mo	Unlimited	gpt5-chat, visual-crop-analysis, geo-consumption-analytics

6. Node-RED Integration: Live Distribution

The node-red-contrib-leafengines module is now publicly available for installation across three platforms:

6.1 Distribution Channels

Node-RED Flows Library: https://flows.nodered.org/node/node-red-contrib-leafengines

npm Registry: https://www.npmjs.com/package/node-red-contrib-leafengines

GitHub (Source & Docs): https://github.com/QWarranto/node-red-contrib-leafengines

6.2 Installation

```
npm install node-red-contrib-leafengines
```

Or install via the Node-RED Palette Manager: search for 'leafengines' and click Install.

6.3 Architecture

Each LeafEngines endpoint maps to a dedicated Node-RED node with typed inputs/outputs, built-in credential management, and automatic tier-gating. Nodes degrade gracefully when offline, serving cached data with staleness indicators.

- Drag-and-drop soil, water, and crop intelligence into any flow
- API key authentication via Node-RED credential store
- Tier-gated nodes surface upsell opportunities organically
- Compatible with Node-RED Dashboard 2.0 for real-time visualization

7. Performance Benchmarks

Server-side latency measured at the Supabase Edge Function layer (excluding network transit):

Endpoint	P50	P95	P99
`get-soil-data`	18ms	42ms	67ms
`county-lookup`	12ms	28ms	45ms
`agricultural-intelligence`	85ms	180ms	320ms
`carbon-credit-calculator`	45ms	95ms	150ms
`gpt5-chat`	250ms	800ms	1.2s

8. Security & Compliance

- API keys stored as SHA-256 hashes with system salt; plaintext never persisted
- Row-Level Security (RLS) on all tables; users access only their own data
- SOC 2 Type II alignment with continuous compliance monitoring
- GDPR-ready: data classification, DPIA templates, SAR procedures documented
- Patent protection: U.S. Patent Applications #19/320,727 and #19/544,827 cover proprietary positioning and sensor fusion algorithms

9. Contributor Program

LeafEngines operates a structured open-source contributor program across four tracks:

Integration Track: Build Node-RED nodes, MCP tools, or third-party connectors

Data Track: Contribute regional soil datasets, calibration data, or validation benchmarks

Domain Track: Provide agricultural domain expertise, crop models, or regional growing guides

Security Track: Responsible disclosure, penetration testing, and security audits

Contributors earn Founders Series API keys with lifetime pricing locked at early-adopter rates. Top contributors are featured in the LeafEngines documentation and podcast.

9.1 Getting Started

1. Get your free sandbox API key at soilsidekickpro.com/podcast
2. Install node-red-contrib-leafengines from npm or the Palette Manager
3. Build an integration and submit via GitHub
4. Earn your Founders Series key

10. Get Started

Get Your Free Sandbox API Key

soilsidekickpro.com/podcast

Download & Install:

flows.nodered.org/node/node-red-contrib-leafengines
npmjs.com/package/node-red-contrib-leafengines
github.com/QWarranto/node-red-contrib-leafengines

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U.S. Patent Applications #19/320,727 & #19/544,827